Ricardo (Zhicong) Xie

Simulation / Analysis Intern | FEA, CFD and Validation

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Valid NZ student visa | No sponsorship required | 25h/week during semester | Full-time 17 Nov 2026 - late Feb 2027

PROFILE

- Mechanical master's student focused on structural FEA, thermal-fluid CFD and model credibility checks.
- Deliverables include load cases, boundary conditions, mesh checks, balance review and model limits.
- Project screenshots and source links: https://zxie588-alt.github.io

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present
Current study and project work in building services, smart manufacturing, CFD/thermal modelling, composite structures and engineering methods.
Relevant coursework: Finite Element Analysis, Computational Fluid Dynamics, Composite Materials, Thermal Modelling

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025
Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PRACTICAL AND TEAM EXPERIENCE

- Manufacturing workshop training: prepared AutoCAD/DXF laser-cut profiles and compared drawings with cut parts.
- Engineering practice: interpreted force-displacement tests, controlled a robotic arm and led a small automotive-structure walkthrough.
- Technical communication: former student news lead with photography and project-documentation experience.

PROJECT WORK

Composite Formula SAE Front Wing Analysis
Abaqus/Standard, shell FEA, composite layup, structural checks
- Built an Abaqus shell FEA model with S4R elements, laminate layups and local mesh refinement.
- Converted aero, lateral cornering and 4.1 G bump conditions into structural load cases.
- Checked ply-level stress use; peak utilisation was about 74% under the 4.1 G bump case.

Thermal-Fluid Simulation & Validation - Solar Chimney Case Study
ANSYS Fluent/CFX, conjugate heat transfer, natural convection
- Built a CFD validation workflow for a laboratory-scale solar chimney with coupled air, absorber and cover domains.
- Checked mesh sensitivity, mass/heat balance, Reynolds/Rayleigh scaling and solver behaviour.
- Documented model boundaries, limitations and recommended physical data points needed for future experimental correlation.

Ruggedized Outdoor IoT Sensor Module Design Package
SolidWorks, enclosure design, DFM notes, Keil C, validation workflow
- Designed a SolidWorks enclosure with gasket stack-up, screw bosses, PCB envelope and service access.
- Prepared DFM notes for CNC review, 3D printing and assembly tolerance checks.
- Screened corner-load, strap-lug and screw-boss regions with an FEA-style check.
- Built Keil C firmware structure with sensor, battery and state-machine modules.

Small Residential MVHR / ERV Revit MEP Coordination
Revit 2026 MEP coordination, MVHR/ERV layout, airflow schedule, SolidWorks, AutoCAD/DXF, NZBC G4/H1 context, commissioning notes
- Built a NZ residential MVHR/ERV package with Revit MEP, CAD evidence and review notes.
- Modelled MVHR unit, ducts, terminals and hangers in SolidWorks; exported STEP, STL, DXF and PNG.
- Created a Revit 2026 model with plan, 3D view, airflow schedule, sheets and simple clash notes.
- Checked 60 L/s supply, 45 L/s extract and 85 L/s boost against duct velocity and pressure allowances.
- Added M-102 pressure review, BIM issue log and Mitsubishi Lossnay LGH-35RVX3-E selection note.
- Linked the design to NZBC G4/H1, Healthy Homes context and a commissioning test plan.

Team Systems Engineering Feasibility Study
Stakeholders, requirements, ConOps, trade studies, risk, verification
- Led systems-engineering structure for a 5-person technical feasibility study.
- Built stakeholder map, requirements hierarchy, ConOps, trade criteria and risk register.
- Used verification gates to keep orbit, power, thermal, communications and cost assumptions aligned.

TECHNICAL SKILLS

- Proficient in: Abaqus/Standard, ANSYS Fluent/CFX, shell FEA, CFD, conjugate heat transfer and natural-convection modelling
- Proficient in: load-case definition, boundary conditions, mesh-sensitivity checks, assumptions log, result interpretation and model-limit documentation
- Familiar with: composite failure criteria, ply utilisation checks, Rayleigh/Reynolds scaling, global mass/heat balance validation